

Team Dynamics and Efficiency in Virtual Reality Tank Simulations

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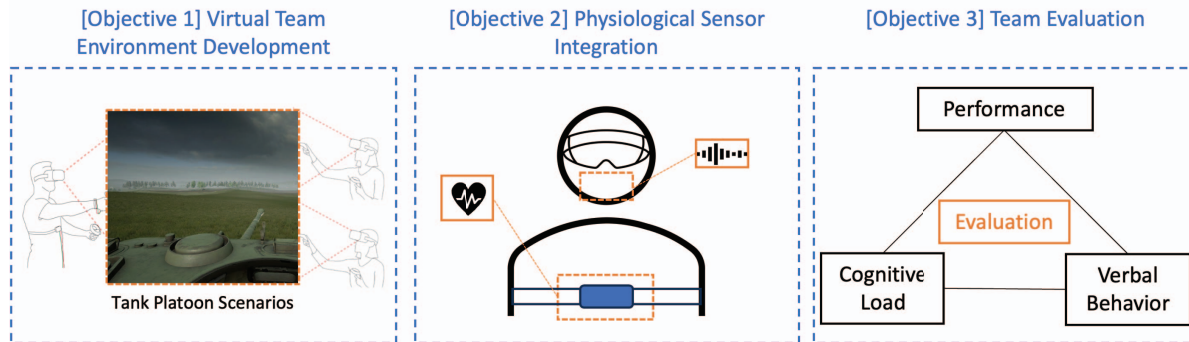


Figure 1: Overview of our research approach, encompassing the design of a virtual teaming environment, integration of physiological metrics, and evaluation of teaming outcomes in a VR tank simulation.

ABSTRACT

This project addresses the pressing need for improved training methodologies for M1 Armor Crewmen in the U.S. Army and Marine Corps, motivated by a significant history of non-combat accidents. Utilizing virtual reality (VR) technology, we adapted a commercially available tank simulation game, *Steel Crew*, to investigate the impact of cognitive load and stress on team dynamics and performance. The simulation varies task difficulty by altering the number of enemy tanks, targeting different operator roles: gunner, commander, and driver. Employing Polar HRV sensors to measure physiological responses, we aim to correlate these with perceived stress and cognitive load. Our hypothesis suggests that increased task difficulty elevates cognitive load and arousal in all roles, with role-specific variations anticipated. This research is pivotal in shaping effective, customized training programs for tank operators, aiming to enhance both technical proficiency and emotional resilience in high-pressure situations.

Index Terms: Human-centered computing—Collaborative and social computing—Empirical studies in collaborative and social computing—

1 INTRODUCTION

The evolution of military training methodologies, especially for high-risk roles like M1 Armor Crewmen, is crucial for operational effectiveness and safety. Historically, the U.S. military has faced challenges in training due to the unique demands of tank operation, such as managing complex machinery and high-stress combat scenarios in confined spaces [14, 16, 18]. To address these challenges, recent advancements in virtual reality (VR) technology have opened new avenues for immersive and practical training [11].

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The application of VR in military training is grounded in its ability to simulate realistic combat environments, providing an interactive platform for skill development under various scenarios [17]. This technology facilitates the creation of diverse training modules, each targeting specific aspects of tank operation, from navigation to decision-making under pressure [9]. VR's adaptability makes it an ideal tool for addressing the inconsistency in advanced training experiences across military units, a factor that has historically contributed to non-combat accidents [15].

To continue, VR training goes beyond skill development to target cognitive and emotional aspects critical to combat operations. Studies have shown that VR can substantially enhance cognitive resource management and reduce stress-related resilience [19]. By coupling VR training with stress-tracking sensors such as Heart Rate Variability (HRV) sensors, we can gather insights into the physiological responses of the military operators while engaging in different training tasks [6].

These comprehensive training programs help prepare service personnel for the field, resulting in fewer non-combat accidents and increased mission success. The findings of this research hold implications for military training but also point to broader uses of VR in other high-risk professions [4, 12]. An overview of this work is presented in Figure 1.

2 RESEARCH APPROACH

2.1 Participants

We plan to recruit 36 participants, forming 12 teams of three to ensure a counter-balanced design in the experimental protocol. The participants within each team can choose their role, reflecting actual military scenarios where crew members often have expertise in specific positions, thereby ensuring more realistic and engaged participation in the simulation. This includes positions such as gunner, commander, and driver. The recruitment process utilizes channels affiliated with Arizona State University's military-related organizations, as their affiliation suggests a degree of alignment in terms of interest and potential exposure to military culture. These recruitment methods include the distribution of flyers, email, and postings on university platforms. To ensure the sample of participants is representative and inclusive, we will ensure a gender-balanced, age-diverse, experience-diverse participant pool. This balance aims to

reflect the evolving dynamics of military personnel and explore any potential demographic-specific responses in training.

Participants undergo a background survey to gauge their prior experience with VR systems and gaming. This survey helps in understanding the baseline level of familiarity and skill among the participants, which is crucial for analyzing the impact of VR training and the learning curve involved.

2.2 Virtual Reality Environment

This study uses Oculus Quest 2 and Pro headsets as the VR head-mounted display. The software chosen is “Steel Crew,” a game offering a realistic tank vs. tank combat experience. The choice of Steel Crew for our virtual environment is supported by other previous successful tank applications in similar military training contexts, thereby enhancing the ecological validity of our research [8]. The game was designed to support various input methods, including gamepads and keyboards, catering to users with different preferences and ensuring wider accessibility [2]. A notable feature of Steel Crew is its detailed tank interiors, crafted to reproduce the tanks as accurately as possible (see Figure 2). This attention to detail in design plays a critical role in enhancing the realism of the simulation, offering participants an authentic experience of operating a tank [2].

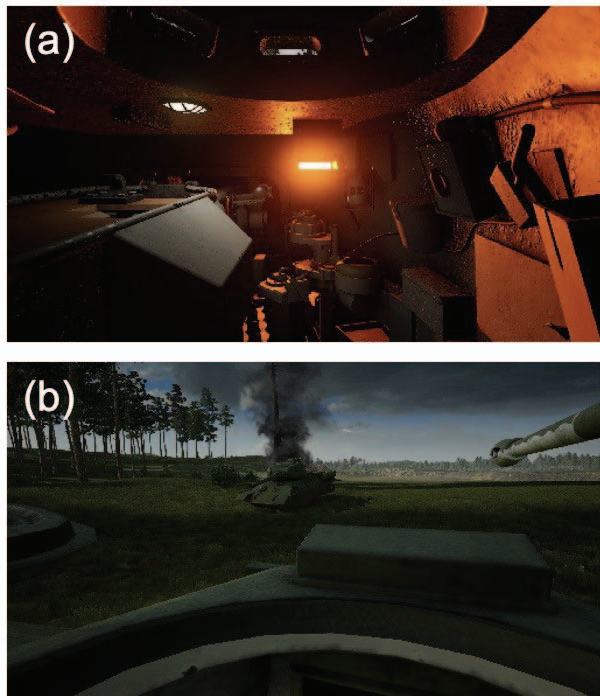


Figure 2: Steel Crew - (a) a realistic replica of the tank interior and (b) an example depicting the role of gunner involved in tank operation [2].

Each role within the tank crew has distinct responsibilities and challenges. The driver, tasked with maneuvering the tank, faces a challenging experience due to limited visibility, further impacted by the position of the hatch—open or closed. On the other hand, the gunner, equipped with a main gun and a coaxial machine gun, must navigate with limited visibility, relying heavily on teammates to identify targets. Their experience is augmented by using motion controllers for precise aiming and control. While not directly involved in controlling the tank, the commander's role is crucial for strategic decision-making. Equipped with tools like maps, monoculars, and the ability to call in airstrikes, the commander must maintain

a comprehensive view of the battlefield and coordinate effectively with other commanders in the team [2].

2.3 Measurements and Data Collection

In our study, we employ physiological, psychological, and performance measures as well as aspects of teaming to evaluate the impact of the VR tank training environment.

2.3.1 Physiological Measures

HRV is monitored using Polar HRV sensors as HRV is a critical indicator of the autonomic nervous system's response. This data provides insights into stress and arousal levels under the VR simulated conditions [1, 3].

2.3.2 Psychological Measures

The participants' stress levels are assessed through self-reported data, collected pre- and post-game using standardized questionnaires like the Perceived Stress Scale [5]. Additionally, cognitive load is evaluated using the NASA-TLX (Task Load Index), which measures perceived workload during task performance [10].

2.3.3 Performance Measures

Several performance measures including the time take to complete specific tasks as well as the accuracy of their actions (i.e., gunner's shooting accuracy) is used to evaluate the effectiveness of the participants in their respective roles.

2.3.4 Teaming Measures

Communication frequency among team members is evaluated based on the frequency and quality of interactions via audio recordings from experimental sessions. Coordination and cooperation are assessed using metrics for evaluating communication styles, planning strategies, and team cohesion, as described by [13] and [7].

2.4 Experimental Protocol

Before each session begins, all Oculus VR devices are charged and sanitized, along with preparing tablets, Polar HRV sensors, and a tripod. Essential items such as consent forms, monetary compensation, and hygiene materials are also arranged.

Upon arrival, participants are seated and their identities are verified. Emphasizing hygiene and wellness, we ask them to use hand sanitizer and conduct a brief wellness check. They are introduced to the VR game where their roles are explained, focusing on the necessity of teamwork in combat scenarios. Participants sign an informed consent form and complete a demographic survey.

Each participant is fitted with a Polar chest strap in a private setting to ensure comfort and accurate data collection. This is followed by a 5-minute baseline recording where participants rest quietly.

A comprehensive briefing on game mechanics and controls is provided, along with a practice session to familiarize participants with the VR environment and their roles. The experiment consists of three sessions, each varying in difficulty. The order of these sessions is counterbalanced between the teams of the experiment. The first session features one ally team (the participants) against no enemies (1 vs. 0). The second session is one enemy team against one ally team (1 vs. 1). The final and most challenging session involves three enemy teams against three ally teams (3 vs. 3), with one of the ally teams being the participants. The tasks in each session were selected based on their increasing levels of complexity and real-world relevance, designed to progressively challenge the participants' decision-making, teamwork, and stress management skills.

After each session, participants complete the NASA-TLX Survey and the Perceived Stress Scale to assess their cognitive load and stress levels. The experiment concludes with a debriefing session, where participants remove their equipment and be provided with

hygiene supplies. Finally, they complete a form to receive their compensation for participating in the study.

2.5 Analytical Methods

We analyze HRV data to assess participants' physiological responses, using statistical measures like mean HRV and frequency domain analysis. Psychological impacts, gauged through the Perceived Stress Scale and NASA-TLX surveys, are evaluated using repeated measures ANOVA and correlation analysis to explore the relationship between stress, cognitive load, and physiological responses.

Performance metrics such as task completion time and shooting accuracy are summarized using descriptive statistics, with a focus on comparing performance across different difficulty levels (1 vs. 0, 1 vs. 1, 3 vs. 3). Communication efficiency and team cohesion data are analyzed qualitatively; communication logs undergo content analysis, while team cohesion responses are subjected to thematic analysis.

Additionally, we plan to use multivariate statistical methods like principal component analysis to uncover patterns in the complex interplay between physiological, psychological, performance, and teaming metrics. This comprehensive analytical approach aims to provide a nuanced understanding of the VR training's impact on individual and team performance, informing future enhancements in training methodologies.

3 DISCUSSION

The upcoming study on the use of VR in military training is set to investigate several key areas, aiming to advance our understanding of how immersive technologies can enhance training methodologies for high-risk roles such as M1 Armor Crewmen. This discussion focuses on the potential implications and considerations for the future study. The study explores how VR's interactive and immersive environment impacts the acquisition of complex skills required for tank operation. Critical to this work is the ability of VR to simulate these high-stress, confined settings that tank crewmen face in combat situations. This involves assessing how participants navigate the VR environment, make decisions under pressure, and adapt to the dynamic nature of virtual combat scenarios.

Within this focus, it is particularly of interest to see the extent to which people with different experiences of VR and gaming respond to the training, as it could help identify the extent to which VR training could be adapted for different demographics. This information could potentially influence how military training programs are designed to accommodate diverse learning styles and backgrounds. A key area of investigation is the impact of VR training on physiological and psychological aspects, such as stress management and cognitive load. By analyzing data from Polar HRV sensors and self-reported stress and workload measures, the study aims to determine how VR training affects the mental and physical readiness of participants. This could have significant implications for understanding the role of VR in preparing soldiers for the psychological demands of real combat.

The study also sheds light on how team dynamics unfold in a VR setting. The interaction between different roles within the tank crew in a virtual environment may provide valuable information on communication efficiency, coordination, and cooperation in high-stress situations. This could be instrumental in developing training programs that enhance team cohesion and collaborative skills, which are crucial in military operations.

Regarding privacy and ethical considerations, all participant data will be anonymized to protect individual identities. Informed consent will be obtained, clearly outlining the nature of the study and the use of data collected. Strict adherence to university and military ethical guidelines will be maintained throughout the study, with special attention to the psychological impact of VR experiences and ensuring participants' well-being during and after the sessions.

4 CONCLUSION

In conclusion, this study aims to provide comprehensive insights into the role of VR in enhancing military training, with the potential to influence training methodologies across various high-risk professions. The study's findings could have implications beyond military training. By demonstrating the effectiveness (or limitations) of VR in a high-risk, high-skill environment, the study could inform the use of VR in other professional fields where training for high-pressure, complex tasks is required. This could pave the way for a broader adoption of VR technologies in various training and educational contexts.

ACKNOWLEDGMENTS

We extend our sincere gratitude to the reviewers for their valuable contributions and insightful feedback on this manuscript.

REFERENCES

- [1] Polar hrv sensors. [<https://www.polar.com/us-en/sensors/h9-heart-rate-sensor>], 2023.
- [2] Steel crew game. [https://store.steampowered.com/app/905970/Steel_Crew/], 2023.
- [3] A. Bayro, Y. Ghasemi, and H. Jeong. A physiological approach to assess arousal in virtual reality and computer-based remote collaboration. In *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*, vol. 66, pp. 2202–2202. SAGE Publications Sage CA: Los Angeles, CA, 2022.
- [4] A. Bayro, B. Havens, and H. Jeong. Xr vest: A novel system for demonstration-based learning of safety skills. *IEEE Transactions on Learning Technologies*, 2023.
- [5] S. Cohen, T. Kamarck, R. Mermelstein, et al. Perceived stress scale. *Measuring stress: A guide for health and social scientists*, 10(2):1–2, 1994.
- [6] R. P. Darken, B. Peterson, and J. Gore. Editorial: Virtual reality applications in military and veteran medicine and healthcare. *Frontiers in Virtual Reality*, 3, 2022. doi: 10.3389/frvir.2022.856666
- [7] M. Demir, N. J. McNeese, and N. J. Cooke. Team communication behaviors of the human-automation teaming. In *2016 IEEE international multi-disciplinary conference on cognitive methods in situation awareness and decision support (CogSIMA)*, pp. 28–34. IEEE, 2016.
- [8] F. Ganier, C. Hoareau, and J. Tisseau. Evaluation of procedural learning transfer from a virtual environment to a real situation: a case study on tank maintenance training. *Ergonomics*, 57(6):828–843, 2014. doi: 10.1080/00140139.2014.899628
- [9] D. J. Harris, T. Arthur, J. Kears, M. Olonilua, E. K. Hassan, T. C. De Burgh, M. R. Wilson, and S. J. Vine. Exploring the role of virtual reality in military decision training. *Frontiers in Virtual Reality*, 4, 2023. doi: 10.3389/frvir.2023.1165030
- [10] S. G. Hart and L. E. Staveland. Development of nasa-tlx (task load index): Results of empirical and theoretical research. In *Advances in psychology*, vol. 52, pp. 139–183. Elsevier, 1988.
- [11] F. Lamberti, F. D. Lorenzis, F. Gabriele Praticco, and M. Migliorini. An immersive virtual reality platform for training cbn operators. In *2021 IEEE 45th Annual Computers, Software, and Applications Conference (COMPSAC)*, pp. 133–137, 2021. doi: 10.1109/COMPSAC51774.2021.00030
- [12] Y. Liu, L. Zhao, and X. Gao. Application of virtual training software in military equipment teaching. *Proceedings of the 2022 3rd International Conference on Mental Health, Education and Human Development (MHEHD 2022)*, 2022. doi: 10.2991/assehr.k.220704.074
- [13] N. J. McNeese, M. Demir, N. J. Cooke, and C. Myers. Teaming with a synthetic teammate: Insights into human-autonomy teaming. *Human factors*, 60(2):262–273, 2018.
- [14] P. Mikołajczak, A. Rychlik, P. Szczyglak, and J. Szuszkiewicz. Diagnostic system simulator of honker vehicle. In *Springer*, 2013.
- [15] R. Qiu, W. Xu, B. Wang, and Q. Shen. A review of research on virtual reality technology based on human-computer interaction in military. In *2022 6th Asian Conference on Artificial Intelligence Technology (ACAIT)*, pp. 1–5, 2022. doi: 10.1109/ACAIT56212.2022.10137916

- [16] N. Sekel, M. Beckner, W. Conkright, et al. Military tactical adaptive decision making during simulated military operational stress is influenced by personality, resilience, aerobic fitness, and neurocognitive function. *Frontiers in Psychology*, 2023.
- [17] L. Soni and A. Kaur. Ar and vr in military combat: Enhancing soldier performance and safety. In *2023 International Conference on Sustainable Computing and Smart Systems (ICSCSS)*, pp. 1205–1209, 2023. doi: 10.1109/ICSCSS57650.2023.10169764
- [18] U.S. Government Accountability Office. Military vehicles: Army and marine corps should take additional actions to mitigate and prevent training accidents. <https://www.gao.gov/products/gao-21-361>, 2021.
- [19] C. Vesa, E. S. oreca˘u, M. S. oreca˘u, and T. A. Vesa. Design, implementation and preliminary testing of a virtual reality system used to train military personnel on a simulated battlefield. *International conference KNOWLEDGE-BASED ORGANIZATION*, 28(3):106–111, 2022. doi: 10.2478/kbo-2022-0094